



PowerShell Conference Europe

Next Up:

Jess Pomfret

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PowerShell Conference Europe

Persist Your PowerShell Script Data with Data API Builder

Jess Pomfret

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jpomfret

Jess Pomfret

(She/Her)

- Data Platform Engineer
- LinkedIn Learning Instructor
- Open Source Contributor
 - dbatools, MicrosoftFabricMgmt
- Passionate about Automation, Proper Football & Fitness

jpomfret7@gmail.com

jesspomfret.com

linkedin.com/in/jpomfret/

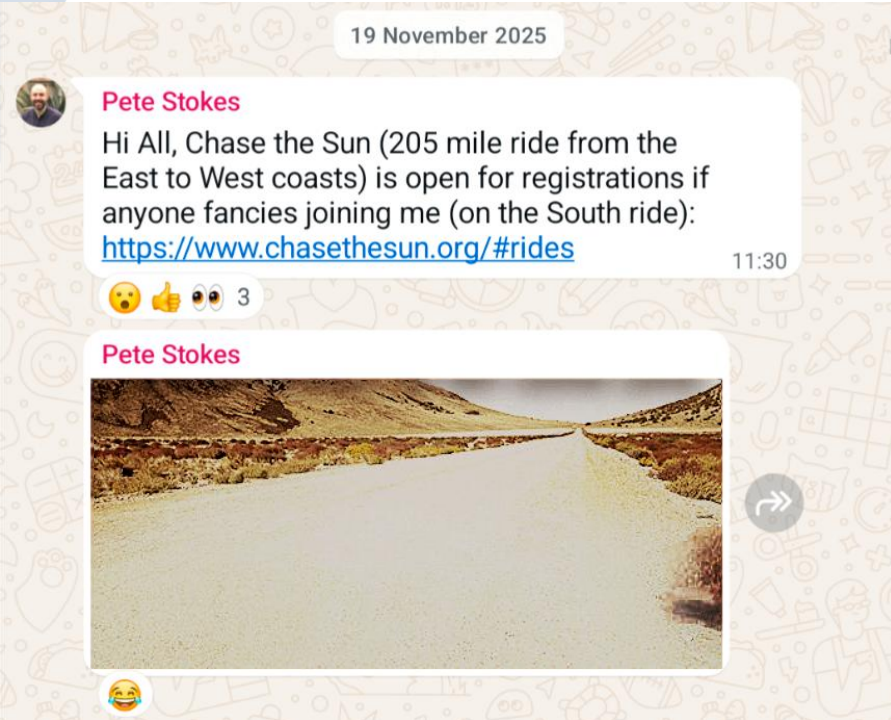


SPEAKER



jpomfret

One day someone asked if I wanted to cycle 205 miles in one day...



**On 20th June 2026,
I'm cycling 205 miles... in one day!**

Chase the Sun 2026

205 miles. Minster-on-Sea to Weston-Super-Mare

Sunrise to sunset. One day. Coast to coast.

3 days after my 40th birthday.



The Problem?

Getting to the start line. Trained. Fuelled. Ready



ret

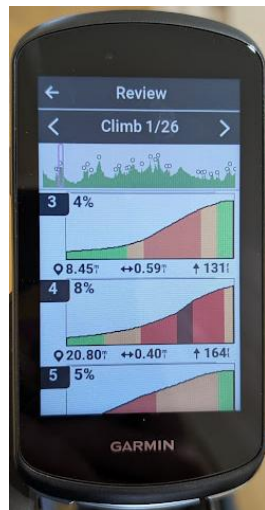
This has generated a lot of data

Wellness, fitness, fueling.

Many devices. Many Apps. No shared view. No way to query it together.

I'm a data nerd. I want it in my database.

No... like A LOT of data



Intervals.icu

Intervals.icu

jpomfret

Activities

Fitness

Power

Pace

Totals

Compare

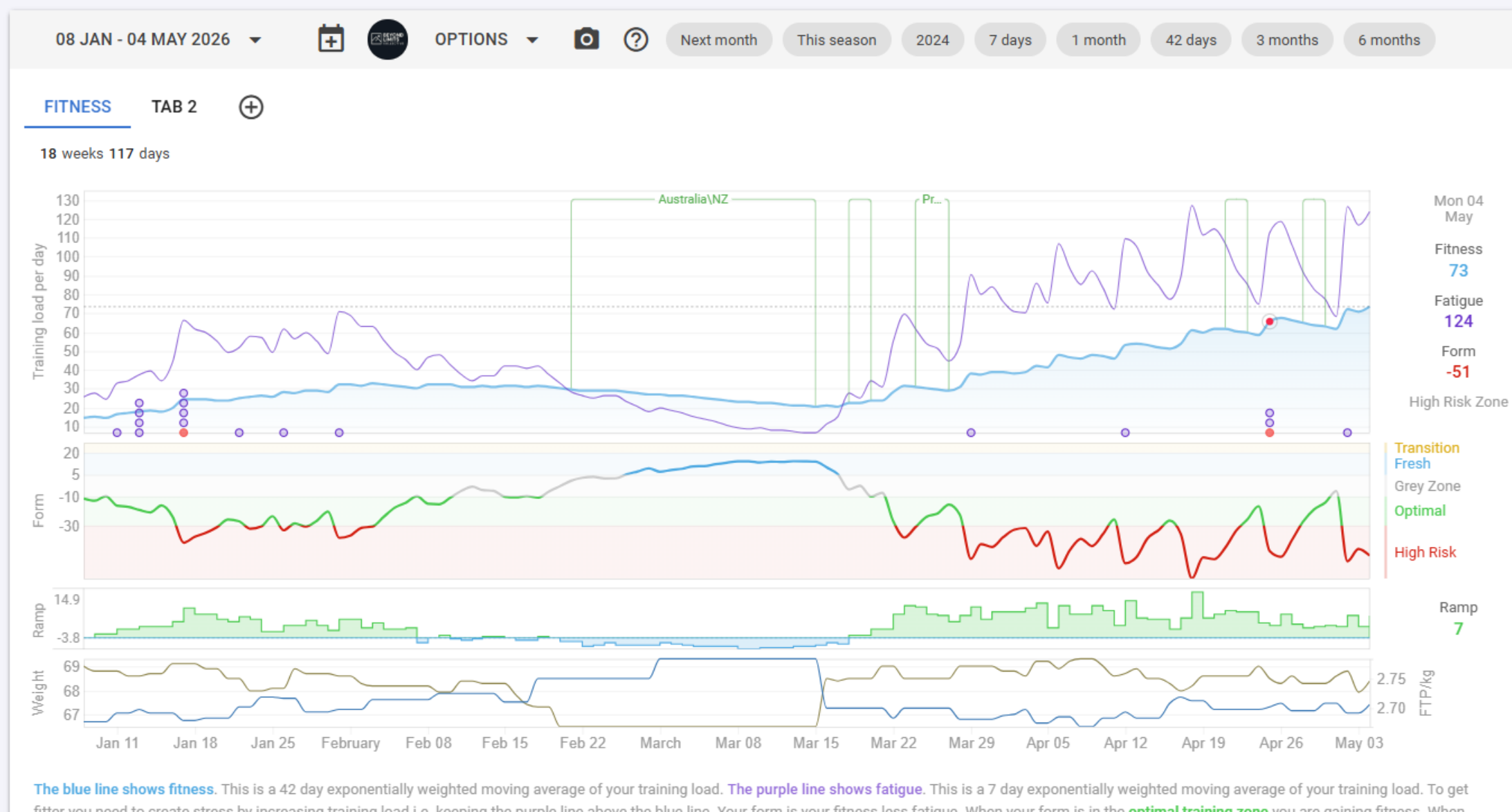
Compare Activities

Settings

Groups

Chats

Forum



FuelGauge

Home grown app

Tracks Calories
Consumed

Compares to Calories
Burnt

Stored in postgres & influx
databases running on Kubernetes
(don't ask)

 **FuelGauge**

Track your daily calorie intake

Logout



Track Calories

View Analytics

← Previous

04 / 05 / 2026

Next →

Today

Copy Yesterday

Breakfast:

285

Lunch:

604

Snack:

0

Dinner:

0

Workout:

928

Bedtime:

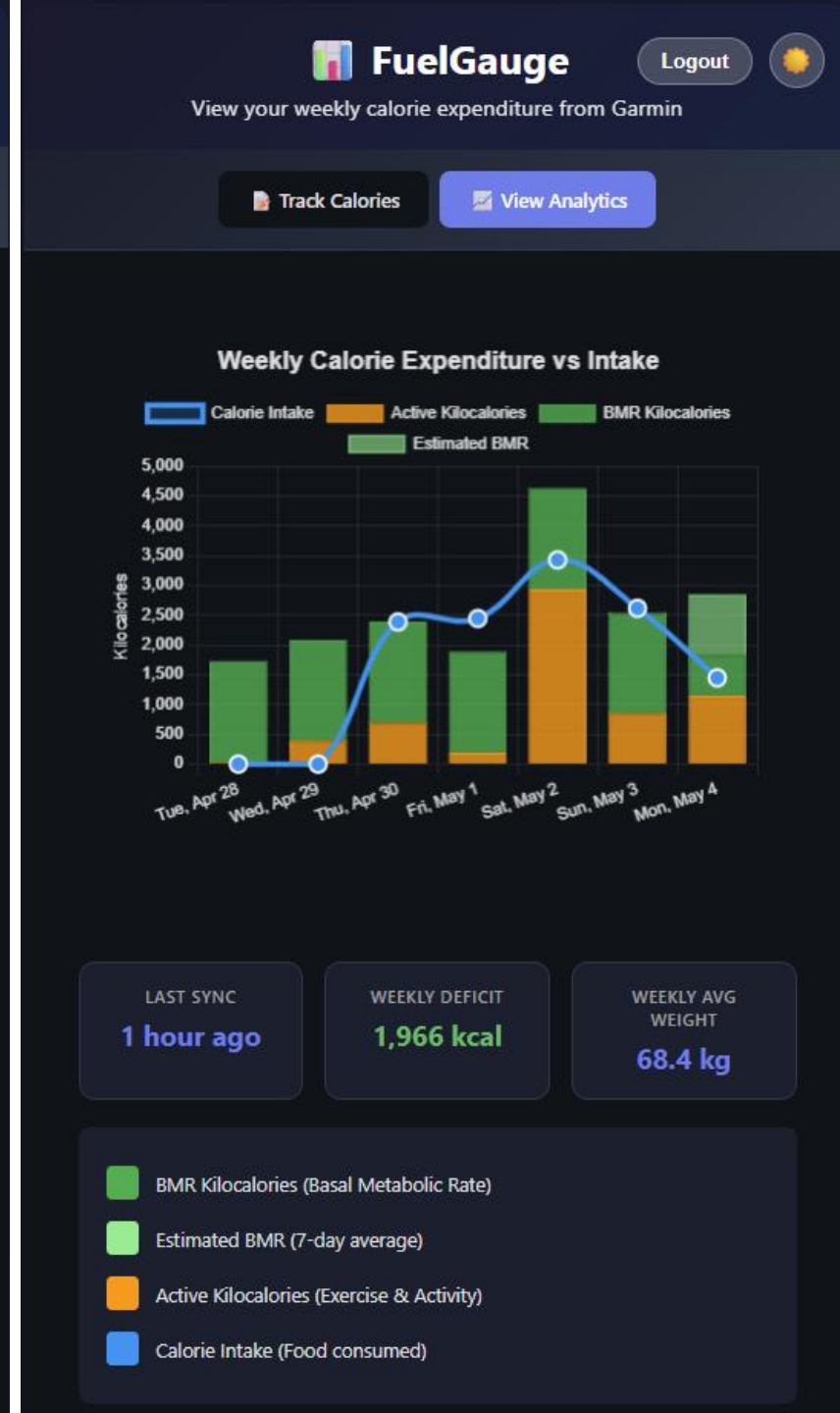
235

Save Entry

Daily Total

2052 calories

789 calories left



One place. One way in. One way out.

Everything in SQL Server.

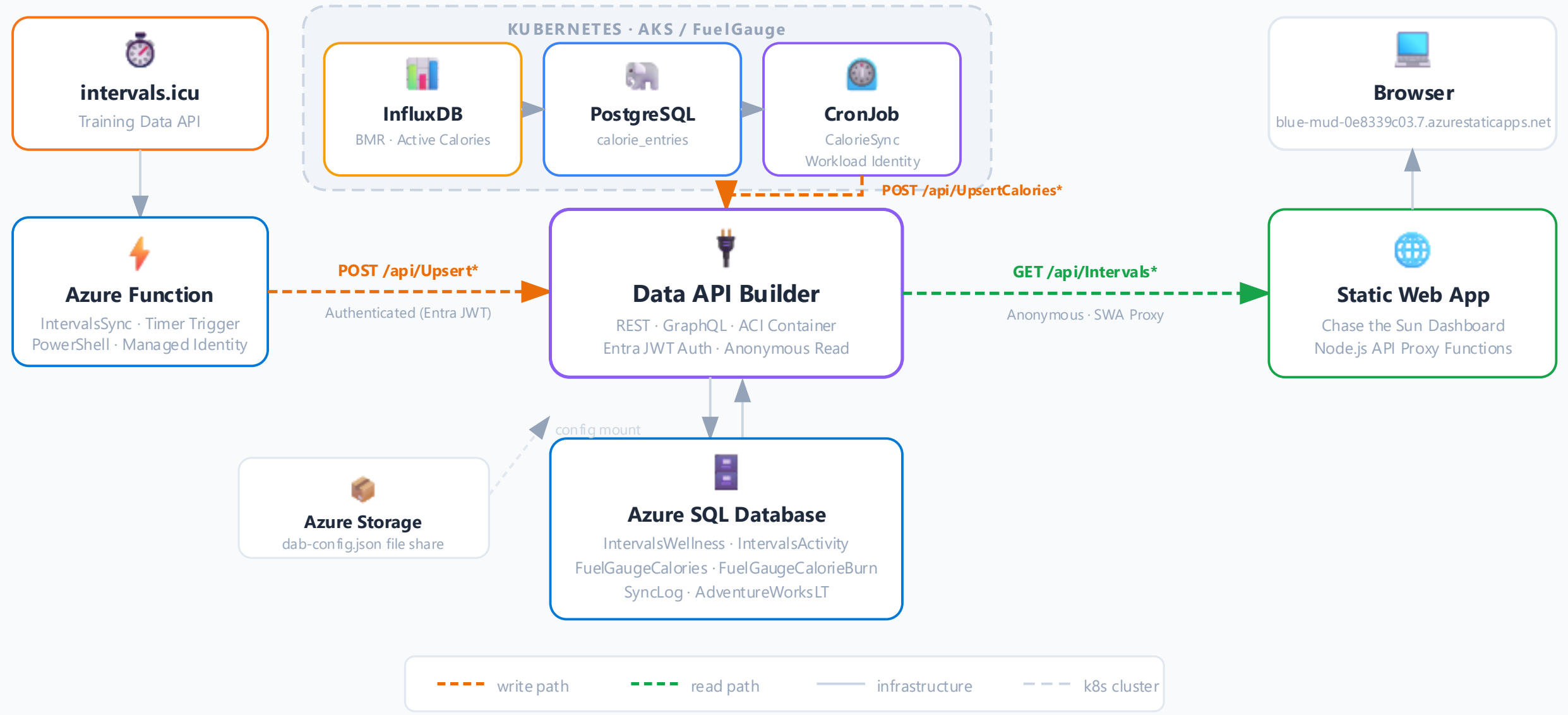
PowerShell writes to it.

A dashboard reads from it.

I do not want to write a backend.

Chase the Sun — Architecture

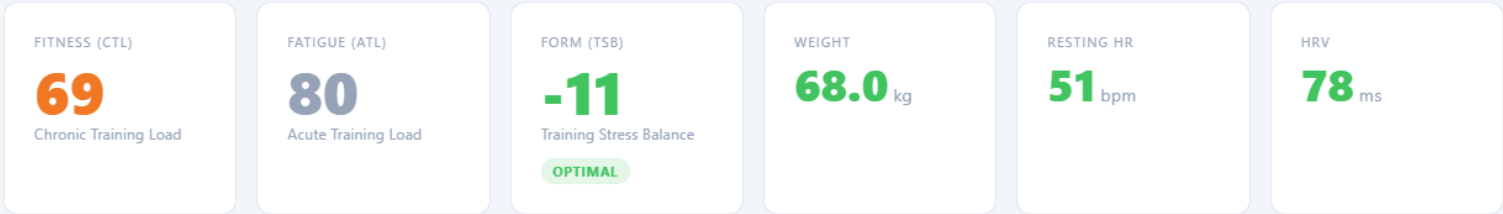
Data API Builder as the central data access layer



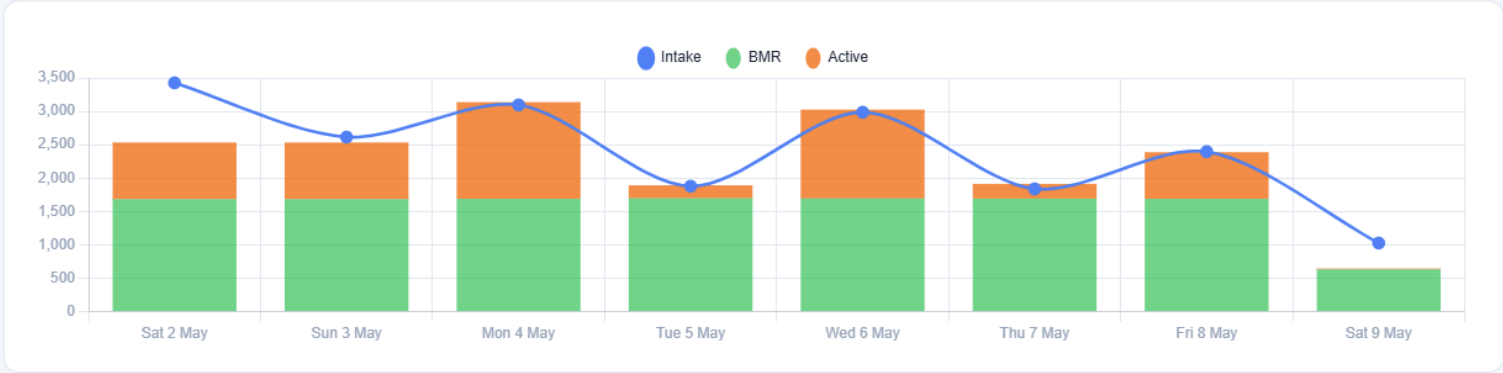
Wellness dashboard

WELLNESS

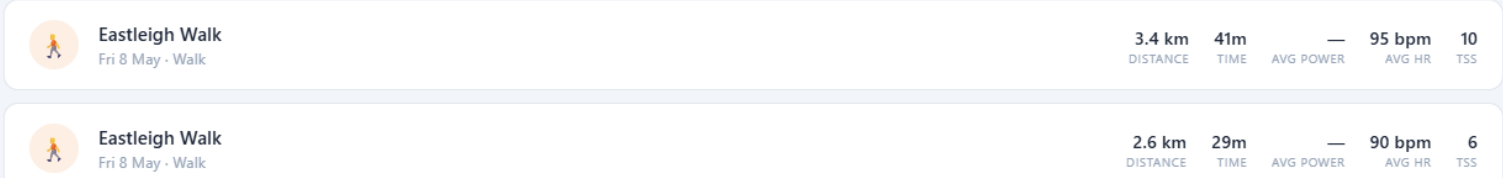
As of Saturday, 9 May 2026



CALORIE INTAKE VS EXPENDITURE

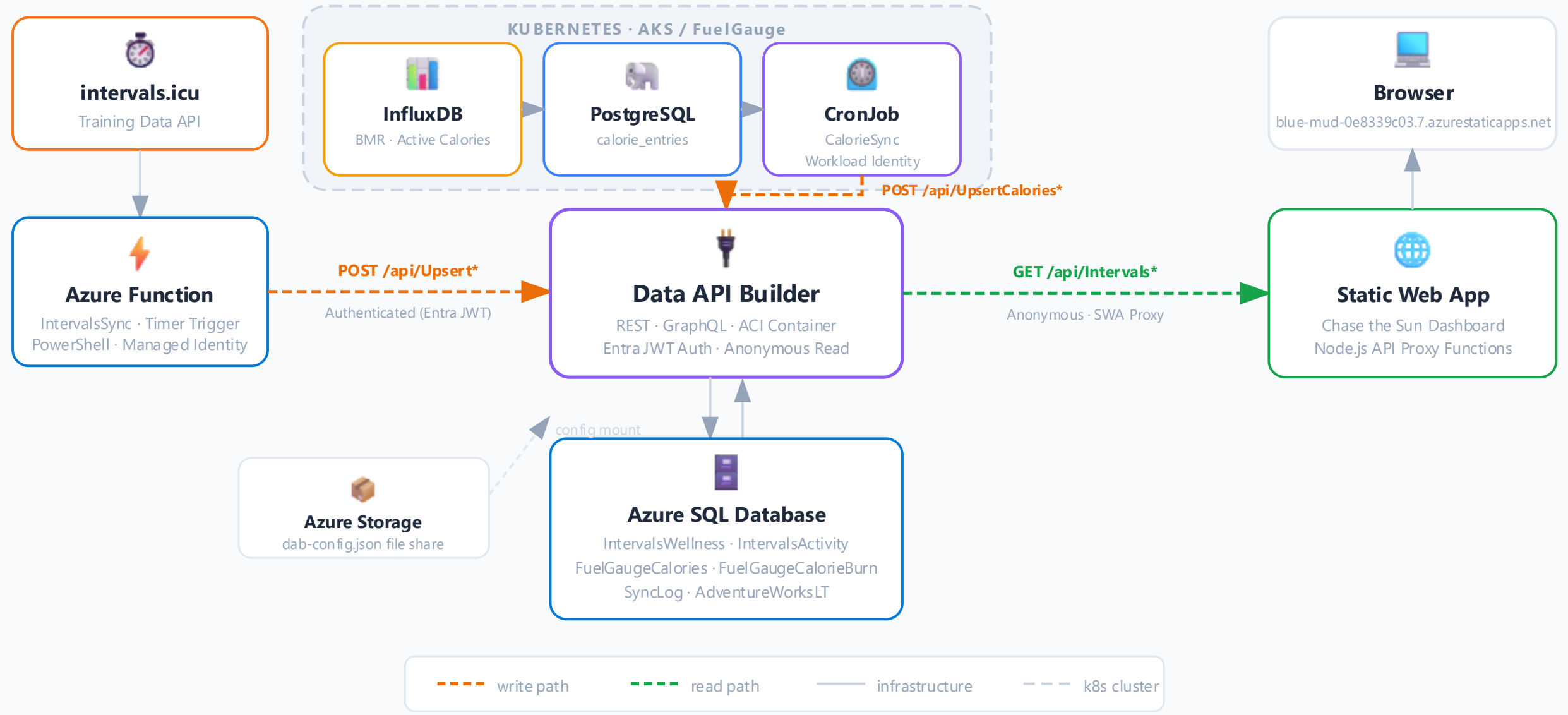


RECENT ACTIVITIES



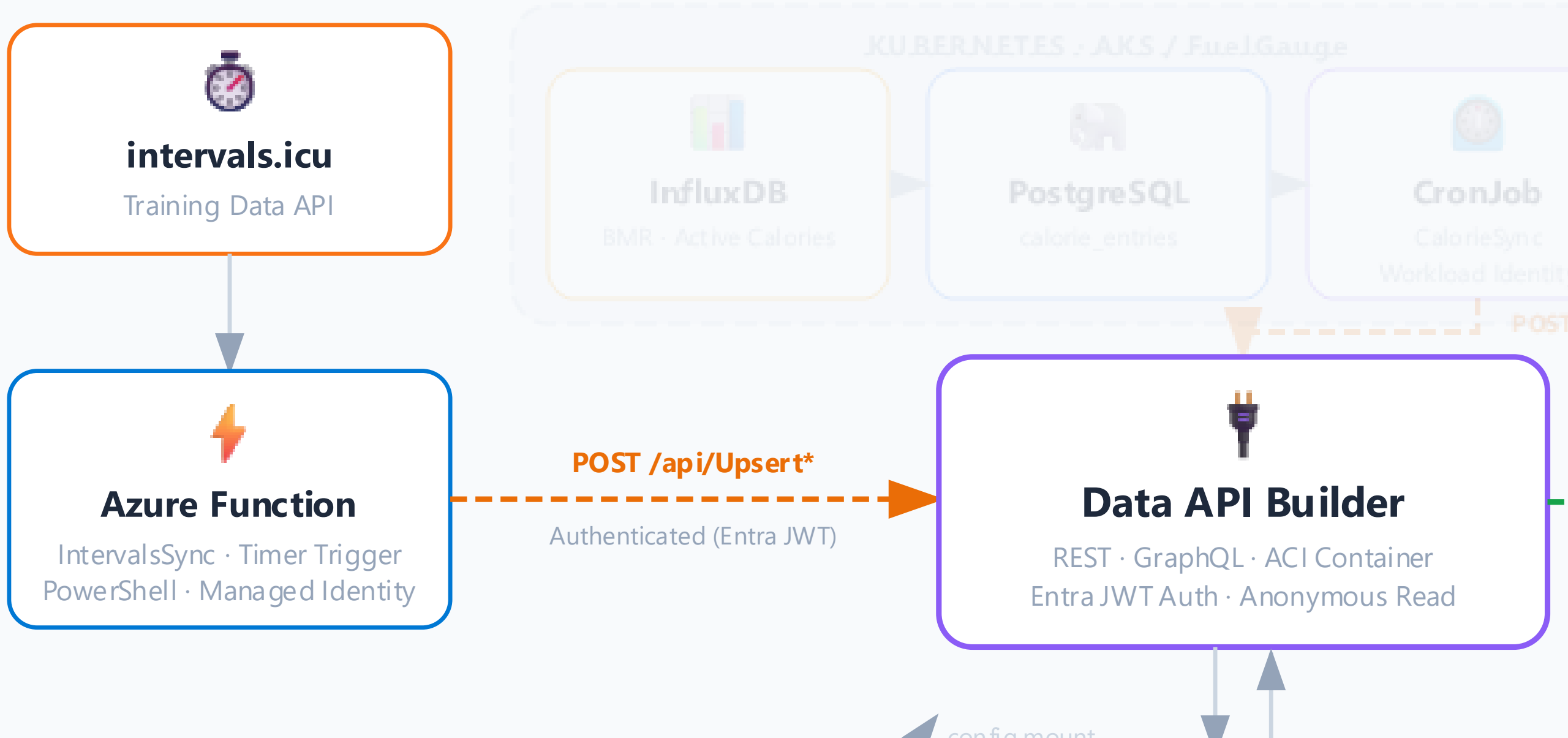
Chase the Sun — Architecture

Data API Builder as the central data access layer



Chase the Sun — Architecture

Data API Builder as the central data access

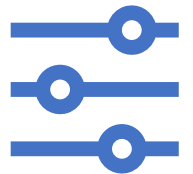


What is Data API Builder?

A config-driven engine that instantly generates REST and GraphQL APIs over your database — with zero backend code.

Just define a **dab-config.json** and you're live.

Open source, Microsoft-backed



Databases & Endpoints

Supported Databases

- SQL Server · PostgreSQL · MySQL · CosmosDB

Expose as HTTP Endpoints

- Tables
- Views
- Stored Procedures



The config file

One JSON file. Two important parts: data source and entities.

```
{
  "data-source": {
    "database-type": "mssql",
    "connection-string": "@env('DATABASE_CONNECTION_STRING')"
  },
  ...
  "entities": {
    "SyncLog": {
      "source": {
        "object": "dbo.SyncLog",
        "type": "table"
      },
      "permissions": [{ "role": "authenticated", "actions": [{"action": "*"}] }]
    }
  }
}
```

Demos

We love Demos!



Deployment

One container image.
Multiple ways to run it.



01

Docker container

Single image, runs anywhere — local dev, CI pipelines, or bare-metal servers. Just point it at your database.

02

Azure Container Apps

Serverless container hosting with auto-scale. No cluster to manage. Pay only for what you use.

03

Azure App Service

Managed PaaS with built-in auth, custom domains, and deployment slots. Familiar to any .NET team.

04

Azure Kubernetes (AKS)

Full orchestration if you already run K8s. Helm chart included. Fits into your existing cluster.

All options use env vars or JSON config — zero code changes between environments

Sorting & Selecting

Sorting — order by any exposed column, ascending or descending

```
GET /api/SyncLog?$orderby=RunAt desc
```

```
GET /api/SyncLog?$orderby=DurationMs asc
```

\$select — slim the payload, fetch only what you need

```
GET /api/SyncLog?$select=LogId,RunAt,Status,DurationMs
```

Filtering

OData-style **\$filter** expressions — query exactly what you need, no extra code.

```
GET /api/SyncLog?$filter=Status eq 'Error'
```

```
GET /api/SyncLog?$filter=Source eq 'IntervalsSync'
```

```
GET /api/SyncLog?$filter=Status eq 'Error' and RunAt ge 2025-01-01T00:00:00Z
```

Operators: eq, ne, gt, lt, ge, le — combine with and / or

Pagination

Cursor-based pagination with **\$first** and **\$after** — no offset, no skipped rows.

GET /api/SyncLog?\$first=10

Response includes nextLink with a \$after cursor token

GET /api/SyncLog?\$first=10&\$after=<token>

Works with any page size — 10, 25, 50 rows — stable under concurrent writes

Authentication

Plug in your identity provider

Entra ID, EasyAuth, any JWT server, or Simulator for dev.

Role-based access control

Anonymous + Authenticated system roles, custom user roles, one effective role per request.

Database policies

Row-level security filter results. Claims flow to SQL session context.

What Can You Persist?

Script Execution Logs — when it ran, result, errors

Automation State — resume where you left off

System Health — inventory, ping results, disk usage

Job Results — pipeline outputs, test data, reports

Anything you'd put in a CSV — now queryable forever.



How It Works



Your function POSTs log data, stats & history to SQL via DAB — zero custom data-access code.

Demos

We love Demos!

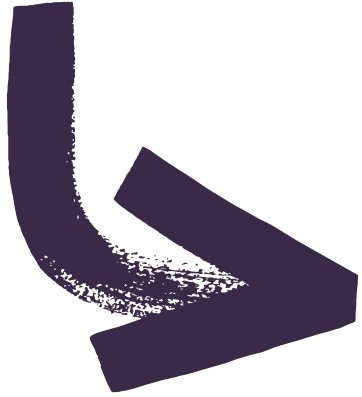


Q&A

15 minutes



GIVE ME FEEDBACK



Persist Your PowerShell Script Data with Data API Builder